Datasheet - Performance test (SFN 3)

General information

Date of evaluation:	July 9, 2025
Reason for evaluation	○ Pre-Check (SFN 1) ○ Process acceptance (SFN 2) ◉ Performance test (SFN 3) ○ Capacity planning
Supplier type	☑ Standard supplier ☐ Subsupplier ☐ JIS/Pearlchain
Supplier to be evaluated:	**China Spring Co., Ltd.**
Part number:	85D 511 305 A/ 85K 511 305A/ 85K 511 305 B
Parts description:	Rear stabilisator

Location

Country:	China
Address (zip, city, street, No.):	201901/ 291 Wenchuan Rd, Shanghai, CHINA
Project manager:	Yangyang, Wang
Supplier number:	
DUNS-No.:	54-530-8561
Name of assessor:	Li Jiajun (C/OP-5)
Department:	Audi China- Supply chain great Asia

Information about the supplier inspection

Date of visit:	2025/7/9~7/11
OEM participants:	Lu, Dongxiao (P/BL) Li Jiajun (C/OP-5)
Supplier participants:	
Required quantity:	4,003 parts/week
Verified capacity: (Pre-Check max. "yellow"!)	4,210 parts/week
Result of capacity evaluation:	**requirements met**

green

Information about the supplier

Employees:	860 employees
Parent company:	SAIC → Hasco-Group → China Spring
Other locations:	Shanghai; Chongqing; Anhui; Tianjin; Business Office in DE & USA
References/ current supply relationships:	Tesla- 20%; VW- 20%

Capacity evaluation - overview

Overview

Part number:	85D 511 305 A/ 85K 511 305A/ 85K 511 305 B
Parts description:	Rear stabilisator
Shifts/week (nominated)	12 Shifts/week

Kapazität: ○ inkl. 15% Flex ◉ exkl. 15% Flex

☐ Start cycle time monitoring

Demands Performance test

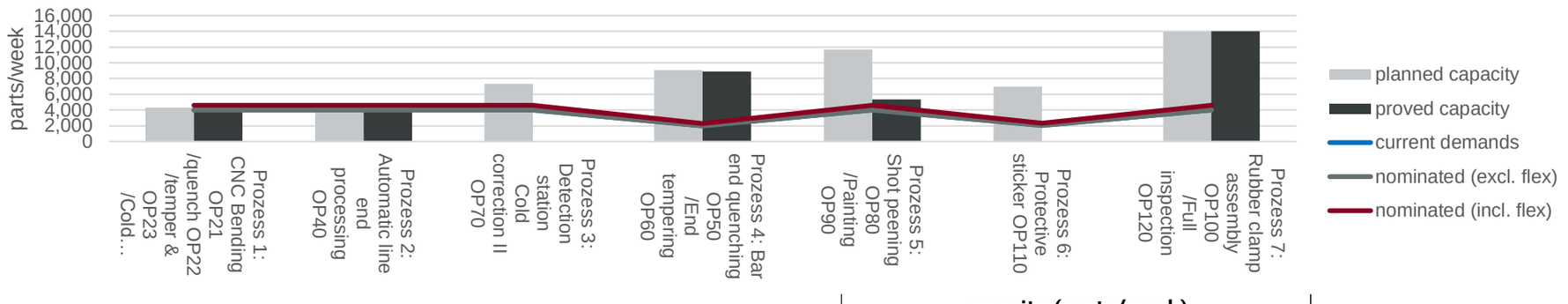
Project	Description	Veh./week	EBR	VR	Net requirement Veh./week	Parts/week	%-share
AU416	85D 511 305 A	2,014	100%	1 parts/car	2,014	2,014	50%
AU511	85K 511 305 A	1,989	70%	1 parts/car	1,392	1,392	35%
AU511	85K 511 305 B	1,989	30%	1 parts/car	597	597	15%
					-	-	0%
					-	-	0%

Overall requirement: 4,003 | 4,003

Nominated total volume

Nominated volume excl. 15% flex	Cars/week:	4,003	Parts/week:	4,003
Nominated volume incl. 15% flex	Cars/week:	4,606	Parts/week:	4,606

Capacity



Process step	Requirement (parts/week)	capacity (parts/week) planned	proved	capacity request
Prozess 1: CNC Bending OP21/quench OP22/temper & OP23	4,003 parts/week	4,283 parts/week	4,268 parts/week	requirements met
Prozess 2: Automatic line end processingOP40	4,003 parts/week	4,178 parts/week	4,210 parts/week	requirements met
Prozess 3: Detection stationCold correction II OP70	4,003 parts/week	7,311 parts/week		
Prozess 4: Bar end quenching OP50/End tempering OP60	1,989 parts/week	9,055 parts/week	8,902 parts/week	requirements met
Prozess 5: Shot peening OP80/Painting OP90	4,003 parts/week	11,698 parts/week	5,355 parts/week	requirements met
Prozess 6: Protective sticker OP110	2,014 parts/week	6,977 parts/week		
Prozess 7: Rubber clamp assembly OP100/Full inspection O	4,003 parts/week	13,926 parts/week	13,989 parts/week	requirements met
Bottleneck capacity	4,003 parts/week	4,178 parts/week	4,210 parts/week	**requirements met**

Remarks

Overview:
The production includes E6 & Q6 two products, and the "Model Change" has be included in "Setup time" record.
Planned system availiablity have been checked seperately. Which results are used for adjust "System availability planned [%]" in <Capacity Calculation> sheet.

Provided capacity: 4,210 parts/week.
PCA3 capacity result of EQ6 & EA6 was passed successfully. (Target: 4,003 parts/week)

Details:
Process1 (CNC Bending OP21 / quench OP22 /temper & OP23 / Cold correction I OP30)
Proven cycle time is 39.6 sec, which cannot match the C.T. in planning (30s).
1) Recheck the production planning of this process, the targeted demand will be reduced by Tesla "Module Y" (From 600tsd/y to 250tsd/y). The (line) capacity occupancy including PPE project is 67%, then **adjust the "Available capacity for project [%]" of PPE from 30%→40%**.
2) Based on the production arrangement at the 1st shift each day, automatic line will not be stopped during 10:30 to 11:00 lunch time.
3) Period2 & Period4 are the unplanned interruptions due to unqualified model changeover activities (some micro stopping were distributed into produciton Period1; Period3 & Period5) (refer to <Capa. record & calculation sheet>), which impection is considered in "Planned system availability" calculation.
***According to above 2) & 3), adjust the "Planned system availability" of Process1 to 83%** (refer to <Planned system availability simulation>), **optimizing of model changeover activities is a high priority.**

Process2 (Automatic line end processing OP40) (The bottleneck of whole process)
The fastest cycle time can reach 40sec but cannot reach 30sec, the tracked C.T. in PCA3 is 41.9sec.
Considering the Process2 is the linked & followed up process of Process1, adjustment of "Available capacity for project [%]" of PPE from 30%→40%" of Process2 is possible and necessary.

Process6 (Protective sticker OP110) The special process of EQ6
No checking of this process due to the necessity has not been determined so far.

Capacity Calculation

Summary

Select following options in QPNI:
"Acceptance w/o non-productive time" & "Calculation flexibility with max. shifts"

Part number:	85D 511 305 A/ 85K 511 305A/ 85K 511 305 B
Parts description:	Rear stabilisator
Shifts/week (nominated)	12 Shifts/week
For evaluation required capacity:	4,003 parts/week
Nominated capacity:	4,003 parts/week

Linked processes

○ Linked processes ● Processes not linked

Distribute the available capacity for OEM equally (acording run@rate) on the marked processes:

Distribute capacities

Hours	Minutes
0.00 h	0 min
Start [hh:mm]	**End [hh:mm]**
0:00	0:00

Production process

	Prozess 1	Prozess 2	Prozess 4	Prozess 5	Prozess 7
General data					
Process name	CNC Bending OP21 /quench OP22 /temper & OP23 /Cold correction I OP30	Automatic line end processing OP40	Bar end quenching OP50 /End tempering OP60	Shot peening OP80 /Painting OP90	Rubber clamp assembly OP100 /Full inspection OP120
Tool designation	**Cold bending tools Resistance heating Tempering furnace**	**Punch press**	**Resistance heating Tempering furnace**	**Shot blasting machine Powder coating**	
Number of (identical) tools		1 tool(s)		1 tool(s)	
Number of cavities				192 cavity(s)	
% of parts handled in this process step [%] (Nomination)	100.00%	100.00%	49.69%	100.00%	100.00%
Remark			**Additional process only for EA6 (AU511 project)**		
% of parts handled in this process step [%] (SFN)					
Established production time					
Shifts/day	2.0 shift(s)	2.0 shift(s)	2.0 shift(s)	2.0 shift(s)	2.0 shift(s)
Shifts/week	12.0 shift(s)	12.0 shift(s)	12.0 shift(s)	12.0 shift(s)	12.0 shift(s)
Minutes/shift (excluding breaks, net working time) [*min*]	675 min	660 min	660 min	660 min	660 min
Hours/shift (excluding breaks) [*h*]	11.25 h	11.00 h	11.00 h	11.00 h	11.00 h
Hours/week (excluding breaks) [*h/week*]	135.00 h/week	132.00 h/week	132.00 h/week	132.00 h/week	132.00 h/week
Process data					
System availability planned [%]	83%	85%	85%	85%	85%
In system availability considered time for breaks per shift [*min*]	45 min	60 min	60 min	60 min	60 min
System availability (**excluding breaks**) [%]	88.53%	92.73%	92.73%	92.73%	92.73%
In total available production time [h/week]	119.52 h/week	122.40 h/week	122.40 h/week	122.40 h/week	122.40 h/week
Available capacity for project [%]	40.00%	40.00%	100.00%	15.00%	100.00%
Available production time for project [h/week]	47.81 h/week	48.96 h/week	122.40 h/week	18.36 h/week	122.40 h/week
Capacity used for other projects [%]	60.00%	60.00%	0.00%	85.00%	0.00%
Production time used for other projects [h/week]	71.71 h/week	73.44 h/week	0.00 h/week	104.04 h/week	0.00 h/week
Run@Rate					
Production time [*min*]	448.00 min	492.00 min	66.00 min	87.00 min	147.00 min
Planned breaks included in production time [min]	0.0 min	0.0 min	0.0 min	0.0 min	0.0 min
Planned interruptions (set up time etc.)	137 min	178 min	0 min	8 min	0 min
Not planned interruptions (downtime)	22.0 min	0.0 min	0.0 min	0.0 min	0.0 min
Production time (net, excl. break/interruptions) [*min*]	289 min	314 min	66 min	79 min	147 min
Produced parts					
ok	430 parts	450 parts	80 parts	384 parts	280 parts
ok with rework	0 parts	0 parts	0 parts	0 parts	0 parts
not ok	8 parts	0 parts	0 parts	0 parts	3 parts
total produced parts	438 parts	450 parts	80 parts	384 parts	283 parts
Scrap [%]	1.83%	0.00%	0.00%	0.00%	1.06%
Availability rate **excluding breaks** (observed)	65%	64%	100%	91%	100%
Performance rate (observed)	86%	96%	93%	101%	96%
Quality rate (observed)	98%	100%	100%	100%	99%
OEE (Overall Equipment Effectiveness) (observed)	54%	61%	93%	92%	95%
Number of employees					
Result capacity					
Cycle time	39.6 sec	41.9 sec	49.5 sec	2,370.0 sec	31.2 sec
Capacity: parts/shift	889 parts/shift	877 parts/shift	742 parts/shift	2,975 parts/shift	1,166 parts/shift
Capacity: parts/week	**4,268 parts/week**	**4,210 parts/week**	**8,902 parts/week**	**5,355 parts/week**	**13,989 parts/week**
Capacity: cars/week	**4,268 cars/week**	**4,210 cars/week**	**8,902 cars/week**	**5,355 cars/week**	**13,989 cars/week**
Deviation to required capacity	6.62%	5.17%	347.55%	33.76%	249.45%
For evaluation required amount of parts	**4,003 parts/week**	**4,003 parts/week**	**1,989 parts/week**	**4,003 parts/week**	**4,003 parts/week**
For evaluation required amount of vehicles	**4,003 cars/week**	**4,003 cars/week**	**1,989 cars/week**	**4,003 cars/week**	**4,003 cars/week**
For nomination required amount of parts	**4,003 parts/week**	**4,003 parts/week**	**1,989 parts/week**	**4,003 parts/week**	**4,003 parts/week**
For nomination required amount of cars	**4,003 cars/week**	**4,003 cars/week**	**1,989 cars/week**	**4,003 cars/week**	**4,003 cars/week**